

A Level Mechanics

Notes · Edexcel A Level Mathematics — Mechanics · Chapter 7: Projectiles

Chapter 7: Projectiles

Why this chapter matters

Projectile motion is the first mechanics topic where splitting a two-dimensional motion into two independent, one-dimensional suvat problems becomes essential, not just a convenient shortcut. You need to be fluent in resolving an initial velocity into horizontal and vertical components, running two suvat sets side by side that share the same time variable t , and switching between them confidently to answer multi-part questions (find the time, then the range, then the landing speed, all within one question). Notation convention: the angle of projection is written θ throughout this chapter, measured from the horizontal. From Chapter 8 onward the same motion is re-expressed using position vectors and the unit vectors $\mathbf{i}$ and $\mathbf{j}$ — projectiles are the natural bridge into that vector treatment, so the habits you build here (resolving into perpendicular components, keeping horizontal and vertical motion separate) carry forward directly.

7.1 Horizontal projection

Concept

A particle projected **horizontally** leaves the point of projection with an initial velocity that has no vertical component at all. Once it is in flight (assuming no air resistance), the only force acting on it is gravity, and gravity acts straight down. Because there is no horizontal force, there is no horizontal acceleration — by Newton's first law, the horizontal velocity simply stays constant for the whole flight. Meanwhile the vertical motion is exactly free fall from rest, unaffected by whatever is happening horizontally.

This is the single most important idea in the whole chapter, so it is worth stating plainly: **every projectile problem is really two ordinary suvat problems running on the same clock** — one horizontal ($a = 0$) and one vertical ($a = g$, downward) — linked only by a shared value of t . Once you accept that the two directions don't interact, the rest of the chapter is just careful bookkeeping.

Key vocabulary

Term	Meaning
Point of projection	The point from which the particle is launched
Horizontal projection	Projection with zero initial vertical velocity
Time of flight, T	The total time the particle is in the air
Range	The horizontal distance travelled before landing

Independence of motions	The horizontal motion (constant velocity) and the vertical motion (constant acceleration g) do not affect one another
Angle of landing	The angle the resultant (combined) velocity makes with the horizontal at the moment of landing

Key formulas

KEY FORMULA

$$\text{Horizontal } (a = 0): x = Ut$$

$$\text{Vertical } (a = g, \text{ from rest}): y = \frac{1}{2}gt^2$$

$$\text{Landing when } y = h: T = \sqrt{2h/g}, \quad R = UT$$

$$v_x = U, \quad v_y = gT, \quad \text{speed} = \sqrt{v_x^2 + v_y^2}, \quad \theta = \arctan(v_y/v_x) \text{ below the horizontal}$$

Here U is the constant horizontal speed, h is the height of the point of projection above the landing point, T is the time of flight, and θ is the angle below the horizontal at which the particle lands.

General method

Before working through the example, notice the pattern you will use every time:

1. Write down separate "Horizontal:" and "Vertical:" statements listing u , v , a , s , t for each direction.
2. Solve the vertical equation first if you are given a height, since that is usually what determines t .
3. Substitute the value of t you find into the horizontal equation to get the range.
4. Combine the horizontal and vertical velocities at landing (using Pythagoras and trigonometry) if speed or direction is asked for.

Worked example

EXAMPLE 7.1

Question. A ball is projected horizontally with speed 15 m s^{-1} from the top of a cliff 20 m above the sea. Find (a) the time of flight, (b) the horizontal distance travelled before the ball hits the sea, (c) the speed and direction of the ball as it hits the sea. Take $g = 9.8 \text{ m s}^{-2}$.

Step 1 (part a) — set up the vertical motion. The ball starts with no vertical velocity, so taking downward as positive: $u = 0$, $a = g = 9.8$, $s = 20$. **M1**

$$s = ut + \frac{1}{2}at^2 \implies 20 = (0)t + \frac{1}{2}(9.8)t^2$$

Step 2 — solve for t . **M1**

$$20 = 4.9t^2 \implies t^2 = 20/4.9 = 4.0816 \implies t = \sqrt{4.0816} = 2.02 \text{ s (3 s.f.)}$$

A1

Step 3 (part b) — use the horizontal motion. Horizontal velocity is unaffected by gravity, so with $u = U = 15$ and $a = 0$: **M1**

$$x = Ut = 15 \times 2.02 = 30.3 \text{ m}$$

A1

Step 4 (part c) — find the vertical speed at landing. Using $v = u + at$ vertically, with the *unrounded* value of t (2.0207...) to avoid losing accuracy: **M1**

$$v_y = 0 + 9.8 \times 2.0207 = 19.8 \text{ m s}^{-1}$$

Step 5 — combine the components. The horizontal component is still $v_x = 15$ (it never changes), so the resultant speed comes from Pythagoras: **M1**

$$\text{speed} = \sqrt{v_x^2 + v_y^2} = \sqrt{15^2 + 19.8^2} = \sqrt{225 + 392.0} = \sqrt{617.0} = 24.8 \text{ m s}^{-1}$$

A1

Step 6 — find the direction. The angle below the horizontal comes from the ratio of the vertical to the horizontal component:

$$\theta = \arctan(19.8/15) = \arctan(1.32) = 52.9^\circ \text{ below the horizontal}$$

A1

Answer: (a) $T \approx 2.02 \text{ s}$. (b) Range $\approx 30.3 \text{ m}$. (c) Speed $\approx 24.8 \text{ m s}^{-1}$ at 52.9° below the horizontal.

Why the "two separate statements" method matters

WHY THE "TWO SEPARATE STATEMENTS" METHOD MATTERS

It is tempting to treat a projectile as one messy two-dimensional problem and try to solve it all at once — this is exactly where most marks are lost. Writing two short headed statements before any substitution, "Horizontal: $u = \dots$, $a = 0$ " and "Vertical: $u = \dots$, $a = g$ ", forces you to commit to which numbers belong to which direction *before* you start calculating. This single habit earns method marks even if the arithmetic later goes wrong, and it is the single most effective way to stop the extremely common error of accidentally putting g into the horizontal equation (or leaving it out of the vertical one).

Common exam mistakes

- Using g in the horizontal suvat equation, or forgetting it in the vertical one.
- Treating the initial vertical speed as unknown rather than zero for a purely horizontal launch.
- Giving the landing angle above the horizontal instead of below it — a projectile falling from a height always lands moving downward as well as forward, so the angle must be measured below the horizontal.
- Rounding the time of flight too early, which then distorts the range and landing speed calculated from it. Carry at least one extra significant figure through intermediate steps, and round only the final answer.

7.2 Projectiles launched at an angle

Concept

When a particle is launched with speed U at an angle θ above the horizontal, the very first step is always the same: resolve U into a horizontal component $U \cos \theta$ and a vertical component $U \sin \theta$.

Do this before anything else, because everything that follows treats these two components completely separately.

- The horizontal component behaves exactly as in §7.1: it stays constant throughout the flight ($a = 0$).
- The vertical component behaves like a particle thrown straight up with initial speed $U \sin \theta$ and deceleration g . It slows down, momentarily stops at the greatest height, and then accelerates back down exactly as it went up.

If (and only if) the particle lands at the **same height** it was launched from, this vertical motion is symmetric: the time taken to rise equals the time taken to fall, so the total time of flight is exactly twice the time taken to reach the greatest height. This symmetry is what gives the two shortcut formulas below — and it is also exactly the assumption that breaks down in §7.3, so it is worth remembering *why* it works, not just that it does.

Key vocabulary

Term	Meaning
Angle of projection, θ	The angle between the initial velocity and the horizontal
Component of velocity	The horizontal ($U \cos \theta$) or vertical ($U \sin \theta$) part of the initial velocity
Time to greatest height	The time at which $v_y = 0$
Greatest height, H	The maximum vertical displacement above the point of projection
Time of flight, T	Total time in the air; equal to $2 \times$ (time to greatest height) only when launch and landing heights are equal
Resultant velocity	The vector sum of the horizontal and vertical velocity components at a given instant

Key formulas

KEY FORMULA

$$\begin{aligned}
 U_x &= U \cos \theta, \quad U_y = U \sin \theta \\
 v_y &= U \sin \theta - gt, \quad t_1 = U \sin \theta / g \text{ (time to greatest height, when } v_y = 0) \\
 H &= U^2 \sin^2 \theta / (2g) \\
 T &= 2U \sin \theta / g \text{ (equal launch and landing heights only)} \\
 R &= U_x T = U^2 \sin(2\theta) / g \text{ (equal launch and landing heights only)}
 \end{aligned}$$

General method

1. Resolve U into $U \cos \theta$ and $U \sin \theta$ immediately — do this once, at the start, and reuse both values throughout the question.
2. For greatest height, use the vertical suvat equation with $v_y = 0$.

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one.

Common exam mistakes

- Swapping $\sin \theta$ and $\cos \theta$ when resolving — a common slip, since θ is always measured from the horizontal, never from the vertical.
- Assuming $T = 2U \sin \theta / g$ even when the question states a different landing height.
- Forgetting that U_x is constant throughout, and mistakenly applying deceleration to it.
- Sign errors in $v_y = U \sin \theta - gt$ once the particle is past its greatest height — v_y becomes negative there, and that is expected, not a mistake.

7.3 The trajectory equation and further projectile problems

Concept

So far, every worked example has used the time of flight T as a stepping stone to find other quantities. But sometimes it is more useful to eliminate time altogether and find a direct relationship between the horizontal distance x travelled and the height y reached. Starting from the horizontal equation $x = U \cos \theta \cdot t$ and the vertical equation $y = U \sin \theta \cdot t - \frac{1}{2}gt^2$, rearrange the first equation to make t the subject, then substitute it into the second. The result is called the **trajectory equation**: a single equation connecting y and x , with no t in it at all.

$$t = x / (U \cos \theta)$$

$$y = U \sin \theta \cdot x / (U \cos \theta) - \frac{1}{2}g(x / (U \cos \theta))^2$$

$$y = x \tan \theta - gx^2 / (2U^2 \cos^2 \theta)$$

This equation is especially useful for problems where the particle lands below its point of projection, above it, or on a sloped surface — situations where "time of flight = $2 \times$ time to greatest height" no longer applies, and solving for t via the symmetric method from §7.2 simply isn't valid.

Taking y as positive **upward** from the point of projection, a landing point below the point of projection has a negative y -value. A slope through the point of projection at angle α has equation $y = x \tan \alpha$ (with a downward slope taken as negative α); solving this simultaneously with the trajectory equation gives the point where the particle lands on the slope.

Throughout this chapter, the standard modelling assumptions apply: the projectile is modelled as a **particle** (its physical size is ignored), **air resistance is neglected**, and **g is constant** at 9.8 m s^{-2} (or 9.81 , as stated in the question) for the whole flight.

Key vocabulary

Term	Meaning
Trajectory equation	The equation of the path traced by the particle, y in terms of x , with t eliminated
Modelling assumptions (projectiles)	No air resistance; particle model; g constant; the launch point taken as the origin
Landing on a slope	The landing point satisfies both the trajectory equation and the

	equation of the slope
Landing at a different height	$y \neq 0$ at landing, so $T \neq 2U \sin \theta / g$ and the trajectory equation (or two separate suvat solves) must be used

General method

1. State your sign convention for y (usually "upward positive from the point of projection") before substituting anything.
2. Substitute the given x or y value into the trajectory equation, keeping θ and U as given.
3. Simplify the trigonometric coefficients ($\tan \theta$ and $\cos^2 \theta$) as decimals to turn the equation into a quadratic in x (or a linear equation in y , depending what is unknown).
4. Solve the quadratic, and discard any root that does not make physical sense (for example, a negative distance).

Worked example

EXAMPLE 7.3

Question. A particle is projected from a point O at the top of a cliff, with speed 24 m s^{-1} at 35° above the horizontal, out over the sea. The particle lands in the sea, 25 m below O . Find the horizontal distance from O at which the particle lands. Take $g = 9.8 \text{ m s}^{-2}$.

Step 1 — set up the trajectory equation. The landing point is 25 m below O , and upward has been taken as positive, so $y = -25$ there. Substitute $U = 24$, $\theta = 35^\circ$: **M1**

$$-25 = x \tan 35^\circ - (9.8 x^2) / (2 \times 24^2 \times \cos^2 35^\circ)$$

A1

Step 2 — simplify the trigonometric coefficients. Using $\tan 35^\circ = 0.7002$ and $\cos^2 35^\circ = 0.6711$, and computing the coefficient of x^2 in one careful step ($2 \times 24^2 \times 0.6711 = 773.1$, so $9.8 \div 773.1 = 0.012679$): **M1**

$$-25 = 0.7002x - 0.012679x^2$$

Step 3 — rearrange into standard quadratic form. Move every term to one side so the x^2 coefficient is positive, which makes the quadratic formula easier to apply correctly: **M1**

$$0.012679x^2 - 0.7002x - 25 = 0$$

A1

Step 4 — solve the quadratic. Using the quadratic formula with $a = 0.012679$, $b = -0.7002$, $c = -25$:

$$x = (0.7002 \pm \sqrt{(0.7002^2 + 4(0.012679)(25))}) / (2(0.012679)) = (0.7002 \pm \sqrt{(0.4903 + 1.2679)}) / 0.025358 = (0.7002 \pm 1.3260) / 0.025358$$

M1

This gives two roots: $x \approx 79.9$ or $x \approx -24.7$. Since the particle moves out to sea, x must be positive, so the negative root is rejected: **A1**

$$x = (0.7002 + 1.3260) / 0.025358 = 79.9 \text{ m (3 s.f.)}$$

Answer: The particle lands approximately 79.9 m from the base of the cliff, measured horizontally from O.

Why this section is tricky

WHY THIS SECTION IS TRICKY

Sections 7.1 and 7.2 both end with "quick" formulas — $T = \sqrt{2h/g}$, $T = 2U \sin \theta/g$, $R = U^2 \sin(2\theta)/g$ — that work only under specific conditions (a purely horizontal launch, or equal launch and landing heights). This section is the first time those shortcuts stop applying, because the particle lands at a different height from where it started. The instinct to reach for a memorised formula is strong, but here it will give a wrong or meaningless answer. Two skills prevent this: being able to **derive** $y = x \tan \theta - gx^2/(2U^2 \cos^2 \theta)$ from the horizontal and vertical suvat equations from first principles (this sometimes appears directly as a "show that" question), and being comfortable **applying** it, including rearranging it in terms of $\tan \theta$ alone using the identity $\sec^2 \theta = 1 + \tan^2 \theta$, which shows up in some harder variants.

Common exam mistakes

- Using $y = +h$ instead of $y = -h$ (or vice versa) for a landing point below the point of projection. Always state your sign convention for y before substituting.
- Dropping the $\cos^2 \theta$ from the denominator (writing $gx^2/(2U^2 \cos \theta)$ instead of $gx^2/(2U^2 \cos^2 \theta)$).
- Arithmetic slips solving the resulting quadratic, or selecting the negative root instead of the physically sensible positive one.
- For "landing on a slope" problems, forgetting that the slope's own equation ($y = x \tan \alpha$) must be solved simultaneously with the trajectory equation, not used as a substitute for it.

Chapter summary

Idea	Key point
Independence of motions	Horizontal: constant velocity ($a = 0$). Vertical: free fall ($a = g$, downward). Always solved as two separate suvat sets sharing the same t .
Horizontal projection	$T = \sqrt{2h/g}$, $R = UT$, landing speed $= \sqrt{U^2 + (gT)^2}$
Components of initial velocity	$U_x = U \cos \theta$, $U_y = U \sin \theta$
Greatest height / time of flight	$H = U^2 \sin^2 \theta / (2g)$, $T = 2U \sin \theta / g$ (equal launch and landing heights only)
Range	$R = U^2 \sin(2\theta) / g$ (equal launch and landing heights only)
Trajectory equation	$y = x \tan \theta - gx^2 / (2U^2 \cos^2 \theta)$; use for unequal heights or landing on a

	slope
Modelling assumptions	Particle model, no air resistance, g constant throughout the flight

Common exam mistakes

- Mixing horizontal and vertical suvat equations — using g where $a = 0$ should apply, or omitting it where it should be used.
- Sign errors with g or height: not stating a clear "positive direction" convention before substituting into vertical suvat or the trajectory equation.
- Forgetting a non-zero initial height, or assuming landing height equals launch height when the question says otherwise.
- Resolving with the wrong angle — swapping $\sin \theta$ and $\cos \theta$, since θ is always measured from the horizontal.
- Quoting $R = U^2 \sin(2\theta)/g$ or $T = 2U \sin \theta/g$ in a question where the landing height differs from the launch height, where neither shortcut is valid.
- Omitting the modelling assumptions (particle, no air resistance, constant g) when a question specifically asks for them.

You should now be able to

- Resolve an initial velocity into horizontal and vertical components using $U \cos \theta$ and $U \sin \theta$.
- Model horizontal projectile motion using independent horizontal (constant velocity) and vertical (free fall) suvat equations.
- Find the time of flight, range, and landing speed/direction for a horizontal projection and for a projection at an angle.
- Find the greatest height and time to reach it for a particle projected at an angle.
- Derive and use the trajectory equation $y = x \tan \theta - gx^2/(2U^2 \cos^2 \theta)$ to solve problems without explicitly finding t .
- Solve projectile problems where the particle lands at a different height from launch, or on an inclined plane.
- State and apply the standard modelling assumptions for projectile motion (particle, no air resistance, g constant).